SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	**ECE**	**Date:**	**13/10/2026**

Code of Conduct

I Sadda Rajeswari(192312140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sadda Rajeswari

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____

Questions

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7- cell pattern. How many channels does a single cell end up with?

Ans: given,

Total channels=channel bandwidth/total bandwidth

=42000*300

=140

Channels per cell = 140 / 7

= 20 channels per cell

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Ans:

= Cluster size × Channels per cell

= 4 × 15

= 60 channels

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Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Ans:

= Frame duration ÷ Number of slots

= 5 ms / 8

= 0.625 ms

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Ans:

= Total bandwidth ÷ Carrier spacing

= 24 MHz / 200 kHz

= 24,000 / 200

= 120 carrier frequencies

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Ans:

Capacity = Number of cells × Calls per cell

= 6 × 250 = 1,500 calls

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Ans:

Active calls = Calls per second × Average call duration in seconds

= 4,000 × (3 × 60)

= 720,000 active calls

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Ans: Bandwidth = Number of channels × Bandwidth per channel

= 150 × 25 kHz = 3,750 kHz = 3.75 MHz

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Ans:

Number of guard bands = 30 − 1 = 29

Total wasted bandwidth = 29 × 40 kHz

= 1,160 kHz = 1.16 MHz

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Ans:

Speed = 90 km/h

Cell diameter = 2 × 1.5 = 3 km

Distance travelled per hour = 90 km

Handoffs = 90 / 3 = 30 handoffs

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?
Ans:

Capacity = Regional centers × Base stations per center × Users per station
= 8 × 40 × 250
= 80,000 users

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Ans:

Total spectrum = 6 MHz = 6,000 kHz

Number of channels = 6,000 / 30

= 200 channels

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Ans:

Traffic = Subscribers × Calls per hour × Call duration in hours

= 1,200 × 4 × (2.5 / 60)

= 200 Erlangs

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Ans: Total bandwidth = 40 MHz = 40,000 kHz

Users = 40,000 / 400

= 100 users

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Ans: Paging rounds = Total cells / Cells per round

= 200 / 4

= 50 rounds

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Ans: Number of roaming subscribers

= 600,000 × 15 / 100

= 90,000 subscribers

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of

9,000 calls per second. What percentage of its capacity is being utilized?

Ans: Utilization = (Current load ÷ Maximum capacity) × 100

= (9,000 / 12,000) × 100

= 75%

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Ans: Total bandwidth = 30 MHz = 30,000 kHz

Number of channels = 30,000 / 200

= 150 channels

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Ans: Successfully completed handoffs

= 2,500 × 99 / 100

= 2,475 successful handoffs

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Ans: Total cluster user capacity

= 7 × 900

= 6,300 users

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Ans: Frame duration ÷ Number of slots

= 4.8 ms / 8

= 0.6 ms

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Ans: Combined cluster user capacity

= 9 × 600

= 5,400 users

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How

many calls are being dropped every second?

Ans: Calls dropped per second

= 60,000 × 0.4 / 100

= 240 calls per second

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Ans:

Path loss = Transmitted power − Received power

= 43 − (−87)

= 130 dB

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Ans:

Total channels = Total spectrum / Channel bandwidth

= 48 MHz / 200 kHz

= 48,000 / 200 = 240 channels

Channels per cell = 240 / 6

= 40 channels per cell

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise

ratio, in dB, does this represent?

Ans:

SNR = Signal power − Noise power

= −85 − (−115)

= −85 + 115

=30dB